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Effect of Exhaust Gas Recirculation on Combustion Efficiency in a Swirl Stabilized Flame With Blends of Natural Gas and Hydrogen

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Reducing the carbon impact of power-generation and industrial gas turbines can be achieved through blending of low-carbon fuels as well as carbon capture. In gas turbines, carbon capture is facilitated by the use of exhaust gas recirculation (EGR), where exhaust gases are mixed with inlet gases to increase carbon dioxide concentrations in the exhaust, which in turn increases the efficiency of carbon capture technologies. However, increasing the diluent fraction of the reactants can decrease flame static stability and lead to flame elongation, potentially affecting combustion efficiency and emissions. In this study, we measure emissions in a single-nozzle model combustor at a range of EGR levels and compositions for blends of natural gas (NG) and hydrogen (H_2) to understand the impact of EGR on combustion efficiency. Different blends of diluents that mimic the effects of EGR are tested with oxygen mole fractions from 21% (no diluent injection) down to 15%. Blends of NG and H_2 are tested with up to 40% H_2 by volume. Flame imaging is used to better understand the connection between EGR, fuel composition, flame stabilization, and combustion efficiency. As EGR level increases, the flame becomes longer and more diffuse. H_2 blending aids flame stabilization and combustion efficiency. The experimental results are complemented by detailed chemical kinetic modeling to identify the changes in reaction pathways that are driven by high levels of diluents and various fuel compositions. We conclude with a discussion of the impact of low combustion efficiency on both cycle efficiency as well as the performance of downstream carbon capture systems.
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Keywords: exhaust gas recirculation, diluents, hydrogen combustion, emissions

1 Introduction

In recent years, there has been a push to reduce the impact caused by greenhouse gas emissions, specifically carbon dioxide (CO_2). In the United States, over a third of CO_2 emissions are emitted by the power generation sector, with approximately 30% of electricity generation coming from the combustion of natural gas (NG) [1,2]. Within this sector, gas turbine engines are one of the key technologies that rely on combustion of NG. To reduce the carbon impact of gas turbines, a transition toward net-zero emissions has been receiving increasing attention. To this end, various technologies and scenarios are being developed. For example, the use of low-carbon intensity fuels, such as hydrogen (H_2), can result in lower CO_2 emissions. Several demonstrations of hydrogen blending into natural gas in power-generation gas turbines have shown that many machines can operate normally with up to 40% hydrogen [3–5]. Alternatively, the deployment of carbon capture and storage (CCS) technologies can reduce the CO_2 emissions of gas turbines running on carbon-bearing fuels such as NG. However, these solutions can pose issues in terms of engine operability, specifically for the combustion process.

The use of these decarbonization strategies in modern gas turbine engines designed for natural gas operation can have detrimental impacts on combustor operability and cycle performance. For example, the use of H_2 fuel can modify the thermoacoustic instability characteristics of a combustor, increase thermal loading, and can potentially lead to flashback issues [6–8]. On the other hand, the use of CCS can impact on the overall efficiency and performance of a plant. To enable CCS as an economically feasible solution, exhaust gas recirculation (EGR) is used. When using EGR, combustion products are recirculated to the engine inlet, which results in an increase in the CO_2 concentration at the outlet of the engine. This increase in CO_2 concentration improves the efficiency of CCS technologies [9–12]. However, the increased fraction of diluents in the oxidizer stream can affect the combustion process.

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The use of EGR can reduce the overall reactivity of the reactants by either affecting the chemical kinetic rates or by changing the thermophysical properties of the mixture in the combustor, with different EGR composition affecting the combustion process in different ways [13–15]. The effects of EGR can result in decreased flame speed, poor flameholding, and changes in dynamic and static stability, which can result in flame blowout depending on the concentration and composition of EGR used [15–21].

The use of both alternative fuels and EGR has been studied for certain operating conditions and injector configurations. Research has shown that the use of H_2 and EGR can by themselves, drastically change the combustion process and their combined effect can have additional effects on the combustion behavior of a system [14,21]. For example, Galeotti et al. [21] considered the improvements of a CO_2-diluted NG flame with a nonpremixed H_2 pilot and how it affects emissions. They showed an improvement in flame stability and CO emissions over baseline NG conditions, demonstrating the potential for decarbonizing gas turbines through combined EGR and H_2. Cycle analysis studies have demonstrated the possibility of benefiting from both H_2 and EGR solutions without sacrificing engine performance [10]. Additionally, some researchers have looked at the economic performance and challenges of H_2 and EGR combined for improving the performance of CCS technologies and have shown desirable performance compared to baseline cases. These benefits can be particularly important for load-following gas turbines where carbon-based fuels can be used while still maintaining low emissions [9,10]. As such, gas turbine decarbonization with EGR and H_2 is a solution that merits further evaluation.

Our previous work showed the effects of various levels and compositions of EGR on a neutrally piloted, swirl-stabilized model combustor operating on NG [15]. Under the conditions tested, the flame becomes elongated and dynamically stable at intermediate oxygen (O_2) levels. However, at the lower O_2 conditions, the flame exhibits what seems to be a nonthermoacoustic instability most likely driven by extinction and reignition behavior. This work was extended to H_2-enriched natural gas flames, where the same conditions were tested with up to 40% H_2 by volume [20]. The addition of H_2 noticeably improved the static and dynamic stability of the lower O_2 conditions, demonstrating the benefits of combining both EGR and H_2 as viable methods for decarbonizing gas turbine engines. Our prior work, however, did not evaluate the emission performance for these test conditions. As the EGR level increases, particularly for low levels of H_2, the flame is quite long. We expect that the combustion efficiency at these conditions may be subpar compared to the baseline cases, which would result in an increase of carbon monoxide (CO) and unburned hydrocarbon (UHC) emissions. The effects on emissions from the combination of H_2 and EGR are different for different species: CO and UHC tend to increase with EGR and decrease with H_2, while the trend for nitric oxide (NO) is a reduction with EGR and an increase with H_2 [18,19,21]. Understanding how the H_2 and EGR interactions can affect combustor emissions is a crucial step toward understanding the viability of these techniques for reducing CO_2 emissions.

The goal of this work is to quantify the impact of dilution level and composition as well as H_2 level on combustion efficiency for a constant flame temperature test matrix. In this study, combustion efficiency is obtained by measuring emissions for stable flame conditions, eliminating the impact of thermoacoustic instabilities on the flame and its emissions. Combustion efficiency is defined as the ratio of the CO_2 level measured in the combustor to the equilibrium CO_2 level calculated using chemical kinetic modeling. Results show that the combustion efficiency is lowest for the high dilution cases. CO emissions are also highest at the lowest oxygen condition tested. The addition of H_2 in the fuel results in a moderate decrease in the CO emissions at each condition tested. Comparison of flame images between cases shows that the time-averaged flame shape has a relationship with the emission outputs, where highly diffuse flames have low efficiency and high CO emissions. Companion chemical kinetic analysis shows agreement with the trends in CO_2 and combustion efficiency, providing insight into the changes in chemical pathways that occur with increasing diluents in the oxidizer and H_2 in the fuel. To interpret experimental results, perfectly stirred reactor (PSR) calculations are performed to elucidate the experimentally determined combustion efficiency and its impact of unburned hydrocarbon emissions. Reaction pathway analysis is used to identify the ways in which the diluents affect the combustion efficiency and resultant pollutant emissions.

2 Experimental Overview

2.1 Experimental Facility. This work is done in an experimental rig equipped with a dual-fuel injector compatible with liquid and gaseous fuel; our previous work with the same test matrix used a different, gaseous-only fuel injector [15,20]. The experimental rig, shown in Fig. 1, is a variable-length combustor that consists of a single-nozzle model combustor with a dual-fuel injector installed in a settling chamber. In this study, only gaseous fuels are used (NG and H_2). The injector consists of a main circuit with swirler vanes and a central pilot circuit inside a centerbody. The rig is operated at atmospheric pressure and the oxidizer mixture (air, CO_2, and N_2) is preheated to 250 °C using an 88 kW electrical heater. The oxidizer mixture is delivered to the settling chamber through a choke plate where it then branches into the main and pilot circuits. Fuel is delivered directly to the injector, with the main and pilot circuits independently controlled. The fuel is mixed with the air in the injector, resulting in a technically premixed flame. The gases (fuel and oxidizer mixture) are metered using thermal mass flow meters. Approximately 5% of the gases are delivered to the pilot circuit with the remaining fraction delivered through the main annular passage. In this study, the combustor is operated with neutral piloting, where both the main flame and the pilot flame have the same equivalence ratio (ϕ). The injector configuration is representative of industrial hardware used in operational gas turbine engines.

Downstream of the injector is an optically accessible quartz combustor with a length of 34 cm. Further downstream is a metallic, variable-length exhaust section where a water-cooled piston can be moved between combustor lengths (L_{comb}) of 68.5 cm and 132 cm. The combustor length is the distance from the injector outlet to the piston. Moving the piston changes the acoustic modes of the combustor, allowing for a survey of acoustic behavior at each operating condition. In the current study, the combustor is initially tested at various combustor lengths for all operating conditions and only one combustor length that has a thermoacoustically stable flame for all operating conditions is used.

A representative stability map for the combustor is shown in Fig. 2; this map was taken at a variety of combustor lengths at the baseline condition (no diluents) for 0% H_2 in NG and 40% H_2 in NG. Figure 2 also shows the frequency in the spectrum with the highest amplitude. At most combustor lengths, this frequency is in the 200–500 Hz range, which is typical for a longitudinal mode of this rig. The rig exhibits instability at the longest combustor lengths, where instability is defined as a pressure fluctuation amplitude above 1% of atmospheric pressure. This mapping was completed for all the operating conditions in this study, and a similar instability map was seen. For this study, the emission sampling process is conducted for a combustor length of approximately 81 cm to avoid thermoacoustic instability at any of the operating conditions.

2.2 Diagnostics and Data Processing. The rig is instrumented with K-type thermocouples to monitor the gas stream temperature ahead of the injector and to maintain the integrity of various components downstream of the flame. Additionally, dynamic pressure measurements are taken at a port on the dump plane of the experiment with a water-cooled dynamic pressure transducer (PCB 113B28); this signal is used to determine the thermoacoustic stability of the combustor. Additional pressure transducers are located in the fuel lines to monitor fluctuations in the fuel. Pressure signals are acquired at 8921 Hz for 8 s and evaluated in the frequency domain using a fast Fourier transform to identify peaks in the signal. The oscillations in the combustor are characterized by determining

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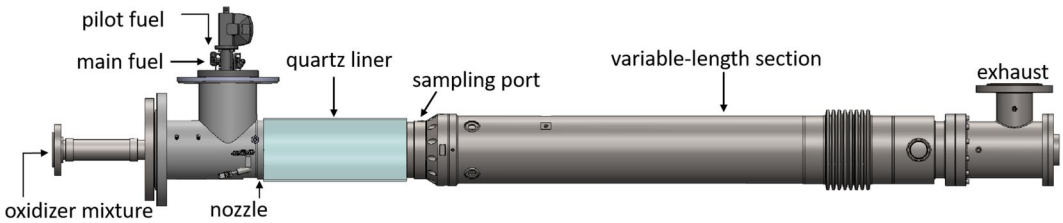

Fig. 1 Experimental combustor rig

the root-mean-squared (RMS) of the acoustic pressure fluctuation at the dump plane (P'_{RMS}) within a ± 2.5 Hz band centered about the oscillation frequency with the highest fluctuation amplitude.

A Photron Fastcam SA4 camera (Tokyo, Japan) was used with a LAVISION intensified relay optics (IRO) intensifier to evaluate the time-averaged flame shape by capturing CH* chemiluminescence images. The camera is filtered with a 430 ± 10 nm bandpass filter. For all test conditions, the flame is imaged at 2000 frames per second over 1 s. For the stable flames evaluated in this study, the acquired image set is sufficient to capture the time-averaged flame shape. The intensifier gain settings are such that the flame is clearly observable without saturating the sensor. The same intensifier gain settings are used for all conditions. The images are time-averaged and an Abel inversion process is used to visualize the center slice of the axisymmetric flame. First, median and averaging filters are applied to the raw images to reduce spatial noise. The flame is considered nominally axisymmetric and so the time-average flame shape is processed using an Abel inversion algorithm by applying a Hankel–Fourier operator [22]. To account for the total heat release rate associated with each pixel location, the images are radially weighted to produce images representative of CH* emission integrated over the azimuthal coordinate.

The emission sampling is done with a Horiba MEXA-584L equipped with oxygen (O_2), CO_2, total hydrocarbon (HC), and carbon monoxide (CO) sensors. The analyzer has a repeatability of 0.01% volume or 1.7% of the reading for CO and 3.3 ppm or 1.7% of the reading for HC [23]. The device was calibrated using prescribed gas mixtures at the start of each test day and verified at the end to measure the drift in calibration. Additionally, the sampling line is cleaned at the end of each test day and HC hang-up tests are conducted on the line before and after testing. A water-cooled, copper-tipped stainless-steel sampling probe is attached to a port downstream of the quartz liner in the postflame region to obtain samples for all test conditions. The probe design has been used in previous studies and has been adequate for measuring emissions in the postflame region [24]. The probe is located 42 cm downstream of the dump plane but 39 cm ahead of the cooled piston, so no water from the piston cooling is expected to be captured by the sampling system. The probe is water-cooled and the gases are delivered to a filter and a water dropout section prior to reaching the analyzer. The cooling water used for the probe is kept at approximately 70 °C. The analyzer and sampling line filters are replaced before every test. The sample has an approximate residence time of 1.4 s inside the probe line before reaching the analyzer, with approximately 0.02 s of residence time in the combustor prior to entering the sampling line. To ensure that the emission measurements are representative of each test condition, the samples are taken only after the flame and the rig temperatures are stable and the analyzer readings have stopped fluctuating. Repeatability tests were conducted for each condition on different days. In addition to the analyzer display values, the measurements are acquired as analog signals directly from the analyzer and sampled over a 16 s window. The acquired signals are time averaged to obtain a representative value for the condition of interest. The time-averaged values are used to verify any discrepancies with the analyzer display values during testing. During testing, the sampling line is inspected to make sure that water is collected in the water dropout portion of the sample line and not in the analyzer inlet or the remainder of the sampling line.

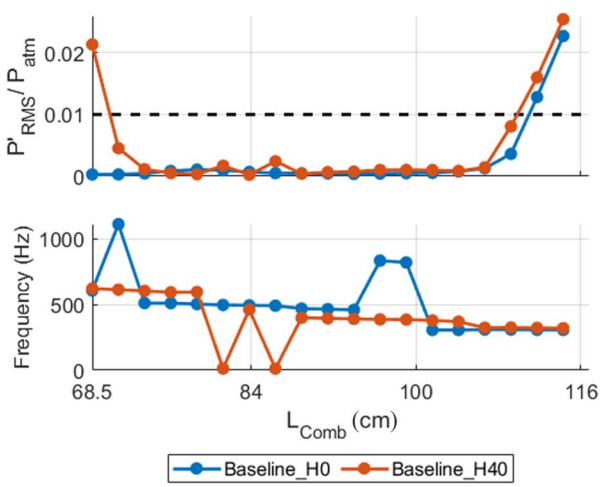

Fig. 2 Representative stability map of the combustor with combustor length for Baseline 0% H_2 and 40% H_2

Before testing in the rig, the analyzer was baselined using a McKenna burner [25] operated with methane and air at room temperature and pressure. The device was tested with a sweep of equivalence ratios from 0.70 to 1.00 as the conditions of interest are lean, premixed flames. The results for the emission sampling, specifically for CO_2, are shown in Fig. 3 as a function of equivalence ratio. The analyzer ideally removes all the water, resulting in a dry basis measurement. The measured data align well with equilibrium values for a dry and wet basis calculated using ANSYS CHEMKIN PRO [26] using USC MECH II [27]. The largest difference between measurements occurred for an equivalence ratio of 0.80 and is less than 3% difference between values, showing good repeatability for the measurements. To account for any variability that the analyzer can introduce, repeat tests will be conducted for all test conditions on different days.

2.3 Chemical Kinetic Modeling. Chemical kinetic modeling with CHEMKIN PRO with the USC MECH II [27] was used to calculate the initial baselining data for the McKenna burner test of the emission sampling device. The equilibrium solver was used to calculate the products for all methane/air conditions tested. Additionally, the test matrix was also designed using the equilibrium solver in CHEMKIN with USC MECH II. The solver was used to calculate the adiabatic flame temperature (T_{ad}) and equilibrium product composition at the inlet conditions (523 K and 1 atm) for each mixture. The adiabatic flame temperature is held constant at roughly 1834 K for the whole test matrix; the equivalence ratio is modified for each condition to maintain that temperature.

While the equilibrium calculation only uses a chemical thermodynamic database, flame behaviors are highly dependent on the fidelity of a chemical kinetic model utilized, which is used to understand the impact of fuel and diluent composition on emissions. To assess the prediction fidelities of chemical kinetic models, four other chemical kinetic models were used: GRI 3.0 [28], REAL FUEL 3

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from Dooley et al. [29], a model from Hashemi et al. (referred to as the DTU model) [30], and the FFCM-2 model from Zhang et al. [31]. Figure 4 compares the calculated adiabatic flame temperatures from equilibrium calculations among five chemical kinetic models. Except for the GRI 3.0 model, four other models predict average adiabatic flame temperatures of 1834 K with a standard deviation of 4 K. The GRI 3.0 model predicts slightly lower than average adiabatic flame temperature at 1830 K with noticeable inconsistencies, which can be attributed to the inconsistency in the thermodynamics database as discussed in Ref. [32].

Figure 5 compares the calculated laminar flame speed (S_L^o) of the premixtures in the test matrix among the five chemical kinetic models. As expected, the predicted laminar flame speed values differ quantitatively among the models. Nevertheless, all models demonstrate qualitatively similar trends depending on fuel/air compositions, along with their Pearson coefficients to each other greater than 0.98. This result suggests that any model can be used for qualitative analysis. Here, we utilized FFCM-2 model from Zhang et al. [31] for chemical analysis, since it is the latest version in the literature, and is well-validated for flame conditions.

Finally, the experiments were performed with natural gas, whereas the key combustion properties are derived from using pure methane as a surrogate for natural gas. The chemical composition of natural gas used in experiments was obtained from the local gas company near Penn State. It is composed of 96.793% CH_4, 3.05% C_2H_6, 0.139% nC_3H_8, 0.007% iC_4H_{10}, 0.011% nC_4H_{10}, thus, can be categorized as a dry natural gas. We computed three key combustion properties: laminar flame speed, extinction stretch rate, and maximum flame speed at the extinction limit. To access the use of methane as a surrogate for dry natural gas, the Pearson coefficients between methane and dry natural gas for these combustion properties are compared and found to be greater than 0.99, confirming the adequacy of methane as a surrogate for dry natural gas.

2.4 Test Matrix. The test matrix for this study focuses on a constant adiabatic flame temperature condition. This test matrix mimics the operational scenario of a gas turbine engine where turbine inlet temperature is maintained as constant. In this case, the adiabatic flame temperature is 1834 K, which is the T_{ad} from a baseline case used in several previous studies [15,20,33–35]. For each diluent and fuel composition, the equivalence ratio is adjusted to maintain the T_{ad} within 11 K of the baseline condition. Table 1 shows the test matrix. For each diluent mixture, the hydrogen content in the fuel varies from 0% to 40% by volume while the O_2 fraction varies from 21% to 15%. The H_2 enrichment of the fuel results in a Wobbe Index range of approximately 48–43 MJ/m^3, with a decrease in the index as the H_2 fraction increases to 40% by volume. The test matrix consists of a Baseline condition with air as the oxidizer along with a set of conditions focused on decreasing the O_2 fraction in the oxidizer stream by adding higher quantities of N_2 (conditions beginning with "N") and a set of CO_2 (conditions beginning with "C"). The naming convention for each test includes the diluent condition, the O_2 content of the oxidizer, and the hydrogen content in the fuel (e.g., N17_H40 is a high-N_2 test condition with 17% O_2 in the oxidizer and 40% H_2 in the fuel).

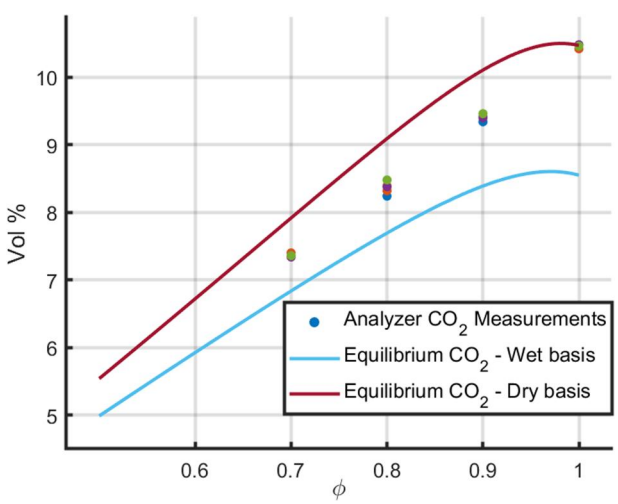

Fig. 3 Comparison of CO_2 measurements and equilibrium calculation values

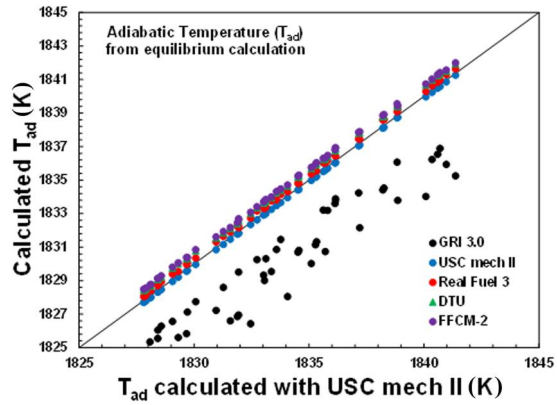

Fig. 4 Comparison of calculated adiabatic flame temperatures from equilibrium calculations among five chemical kinetic models

3 Results and Discussion

3.1 Time-Averaged Flame Shape. Figure 6 shows the time-averaged, Abel-inverted flame shape as a function of both diluent (in columns) and hydrogen level in the fuel (in rows). The top-half of each image is provided at each condition and a sketch of the injector and combustor liner is indicated in the top-left image. The flame has two parts: a main flame, which is stabilized in the inner shear layer of the swirling jet and expands outward until it impinges on the wall, as a typical V-flame, and a central pilot jet flame, which is visible along the centerline in all cases.

The flame has a similar shape for the conditions with an inlet O_2 level above 17%. In all these cases, the main flame is clearly stabilized on the centerbody. The main impact of the decreasing oxygen content is the length of the flame, which increases as the oxygen level decreases. Note that there is not a significant difference between the C17 and N17 cases, meaning that the diluent composition does not have a significant impact on the flame structure at this level of O_2. At the 15% O_2 conditions, the flame is weakly anchored on the centerbody and becomes noticeably

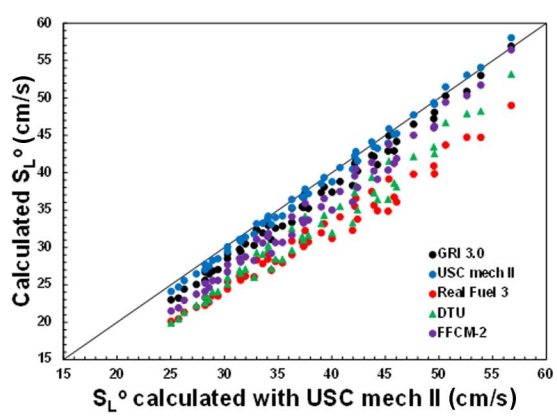

Fig. 5 Comparison of calculated laminar flame speed (S_L^o) of the premixtures in the test matrix among five chemical kinetic models

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elongated and diffuse, especially for the C15 case. This difference between the N15 and C15 conditions could be a result of the thermophysical properties of CO_2, which has a higher specific heat than N_2. Prior work with this test matrix in a different injector showed similar differences in flame structure between the C15 and N15 cases, where the C15 case has a noticeably lower flame speed and extinction strain rate than the N15 mixtures, resulting in a much more diffuse flame.

The effect of the diluents on flame elongation is reduced by the addition of H_2. As the fraction of H_2 in the fuel increases, the flame becomes shorter and more compact. This change is likely a combination of both the higher flame speed of fuel mixtures with hydrogen as well as the higher extinction strain rate, as can be seen in Table 1. For the higher O_2 conditions, the flame has noticeably less activity close to the wall as it becomes more compact and the center of heat release moves closer to the nozzle. For the N15 and C15 conditions, the addition of H_2 reduces the flame length and makes the flame less diffused. At 40% H_2, the N15 and C15 conditions have similar flame shapes and are no longer elongated. The distribution of the flame inside the combustor can be characterized by the use of a coefficient of spatial uniformity as shown in the following equation:

$$\mathrm{CSU} = \sqrt{\frac{\sum (I_i - I_{\mathrm{avg}})^2}{N} * \frac{1}{I_{\mathrm{avg}}}} \tag{1}$$

The coefficient represents how concentrated or localized the flame is in the combustor. A higher value represents a compact flame, while a flame that is evenly distributed will have a lower coefficient. The values of the coefficient, show that the flame becomes more compact at higher O_2 concentrations. As the O_2 fraction decreases, the coefficient decreases, indicating that the flame is more evenly distributed inside the combustor. Increasing the level of H_2 in the fuel decreases coefficient of spatial uniformity at each O_2 condition.

3.2 Emissions Measurements. The combustor emissions, particularly CO, CO_2, O_2, and UHC, were measured at each condition. Figure 7 shows the CO_2 and CO emissions and Fig. 8 shows the O_2 and UHC emissions as a function of inlet oxygen level condition for the three levels of H_2 in the fuel. In the case of CO_2, trends match those predicted from equilibrium calculations, presented in Table 1. For a given O_2 level, the high-CO_2 cases have proportionally higher CO_2 cases than the high-N_2 cases, as was intended in the test matrix. Additionally, the amount of CO_2 at each condition decreases with increasing H_2 in the fuel. These CO_2 results are used in the calculation of combustion efficiency, discussed next.

The CO levels are negligibly small at each of the conditions except for the 15% oxygen conditions. For both C15 and N15, the CO levels are highly elevated compared to the rest of the test matrix. Further, increasing the amount of hydrogen in the fuel decreases the CO fraction, likely a result of both low production of CO and better conversion of CO to CO_2 through routes involving OH. Finally, the unburned hydrocarbon levels are relatively similar across the test matrix. The reason for this lack of trend in UHC is likely twofold. First, the level of unburned hydrocarbons in a lean NG/hydrogen flame is relatively low to start with and likely challenge the fidelity of the instrument at these levels. Further, we expect that some of the heavier UHCs may drop out of the line with the water as it is condensed before the emissions measurement, which would result in a lower level of measured UHC in all cases.

Figure 9 shows the calculated combustion efficiency based on measured CO_2 versus equilibrium CO_2, accounting for the drying process ahead of the emissions measurement. The combustion efficiency is above 0.95 for most of the cases, with a variation between test days of around 0.05 for most conditions, with the exception of the 15% O_2 conditions. The variation presented for the 15% O_2 conditions is independent of the H_2 level and has only been measured in the C15 condition (high CO_2 condition). Our previous work in a similar experiment [15] encountered transient behavior for the C15 condition that was not encountered for the other conditions.

Table 1 Test matrix with H_2 content, oxidizer composition, equivalence ratio, flame properties, and product CO_2

Case	H_2 fuel volume fraction	O_2	CO_2	N_2	ϕ	T_{ad} (K)	S_L^o (cm/s)	$\lvert\kappa_{\mathrm{ext}}\rvert$ (1/s)	$S_{L,\max}$ (cm/s)	Equilibrium CO_2 (% vol. dry)
Baseline_H0	0.00	0.21	0.00	0.79	0.60	1840	45.25	1922	93.76	6.70
Baseline_H10	0.10	0.21	0.00	0.79	0.60	1845	47.96	2494	103.39	6.54
Baseline_H20	0.20	0.21	0.00	0.79	0.59	1834	49.56	2958	114.76	6.23
Baseline_H30	0.30	0.21	0.00	0.79	0.59	1841	53.88	3941	133.45	5.99
Baseline_H40	0.40	0.21	0.00	0.79	0.58	1832	56.72	4951	152.14	5.60
C19_H0	0.00	0.19	0.03	0.78	0.67	1832	40.03	1551	82.15	9.98
C19_H10	0.10	0.19	0.03	0.78	0.67	1837	42.40	2013	92.22	9.82
C19_H20	0.20	0.19	0.03	0.78	0.67	1843	45.41	2328	104.44	9.62
C19_H30	0.30	0.19	0.03	0.78	0.66	1835	47.62	3116	117.69	9.28
C19_H40	0.40	0.19	0.03	0.78	0.66	1844	52.52	4229	138.79	9.00
C17_H0	0.00	0.17	0.08	0.75	0.78	1843	33.54	1149	70.30	15.63
C17_H10	0.10	0.17	0.08	0.75	0.77	1835	34.75	1436	76.57	15.37
C17_H20	0.20	0.17	0.08	0.75	0.77	1841	37.25	1852	87.83	15.18
C17_H30	0.30	0.17	0.08	0.75	0.76	1835	39.29	2327	96.05	14.86
C17_H40	0.40	0.17	0.08	0.75	0.76	1843	43.24	2985	112.84	14.58
N17_H0	0.00	0.17	0.05	0.78	0.76	1835	35.41	1251	73.03	12.22
N17_H10	0.10	0.17	0.05	0.78	0.76	1841	37.38	1548	81.12	12.06
N17_H20	0.20	0.17	0.05	0.78	0.75	1833	39.03	1944	91.36	11.77
N17_H30	0.30	0.17	0.05	0.78	0.75	1840	42.14	2440	101.85	11.54
N17_H40	0.40	0.17	0.05	0.78	0.74	1835	45.33	3209	117.04	11.17
C15_H0	0.00	0.15	0.12	0.73	0.90	1836	25.04	677	51.17	20.04
C15_H10	0.10	0.15	0.12	0.73	0.90	1841	26.61	807	57.11	19.88
C15_H20	0.20	0.15	0.12	0.73	0.89	1836	28.21	1050	63.73	19.61
C15_H30	0.30	0.15	0.12	0.73	0.88	1832	30.30	1319	72.40	19.30
C15_H40	0.40	0.15	0.12	0.73	0.88	1841	34.48	1703	83.11	19.03
N15_H0	0.00	0.15	0.07	0.78	0.87	1834	28.57	786	58.26	14.43
N15_H10	0.10	0.15	0.07	0.78	0.87	1839	30.28	953	63.77	14.26
N15_H20	0.20	0.15	0.07	0.78	0.86	1833	31.95	1231	71.37	13.99
N15_H30	0.30	0.15	0.07	0.78	0.86	1840	34.69	1551	80.90	13.76
N15_H40	0.40	0.15	0.07	0.78	0.85	1837	37.51	2047	93.54	13.40

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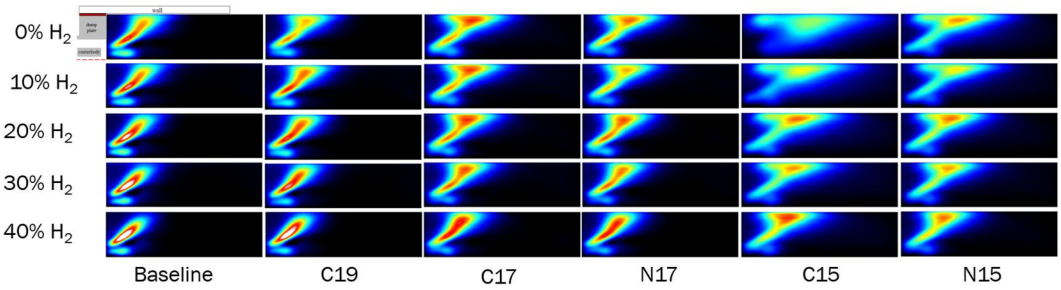

Fig. 6 Time-averaged CH* chemiluminescence imaging of the flame structure

The behavior appeared to be an extinction-reignition behavior in the combustor that was sustained at H_2 levels below 30%. Although the conditions tested here are stable under the 1% P_{atm} criterion, the C15 condition did exhibit some amount of low frequency noise of around 0.4% P_{atm} that could be sufficient to cause variation in the emission measurements. Although there was some amount of noise, the flame itself was not noticeably fluctuating during testing. Another reason for the combustion efficiency behavior of the C15 condition could be its flame length. The condition is more distributed than the other conditions and this may cause greater variation in the measurement of emissions at the exit of the combustor where the emission probe is located.

The coefficient of spatial uniformity is calculated from the time-averaged, Abel-deconvoluted flame shapes to see if there is any relation between the flame shape and the emissions that can be applied for all the diluents and H_2 conditions of interest. The relations obtained for the combustion efficiency and measured CO_2 as a function of the coefficient of spatial uniformity are shown in Fig. 9(*b*). The results for the CO_2 show a clear decrease in emitted CO_2 with increasing coefficient magnitude. The larger coefficient values indicate that the flame is more compact and localized. The compact flame shape tends to happen for conditions with higher flame speeds, which are those with either low levels of diluent injection or sufficiently high H_2 in the fuel, which translates into a reduced CO_2 fraction at the outlet. The highly diluted conditions with low levels of H_2 in the fuel tend to have diffuse and elongated flames that are well distributed within the combustor, leading to a smaller coefficient magnitude. However, the relation in the coefficient of spatial uniformity for the C15 condition follows the behavior of the combustion efficiency and shows some variation for said test case. As previously discussed, this variation could be due to some transient behavior or low amplitude fluctuations in the flame along with the fact the elongated flame might cause greater variation in the emissions at the probe inlet.

3.3 Chemical Kinetic Modeling. Chemical modeling is used to help interpret experimental results as the diluent and fuel compositions change. The use of a chemical reactor network approach is common for interpreting chemical kinetic behaviors in a gas turbine combustor, which requires a substantial optimization process to appropriately estimate the emissions formation comparable to the experimental results [36,37]. While the chemical reactor network will be explored through optimization in the future, here we use a single perfectly stirred reactor (PSR) to assess the role of chemical kinetics on the experimental results from a high level.

Figure 10 shows an example PSR simulation for the baseline condition (Baseline_H40), exhibiting the combustion efficiency defined as the ratio of CO_2 to the CO_2 in equilibrium as a function of flow residence time in the PSR. The combustion efficiency decreases with decreasing flow residence time in PSR. The main UHC at the exit of the PSR are CH_4 and CH_2O and exhibit a rapid increase in their mole fractions as the extinction limit is approached. The solid vertical line indicates the flow residence time between the flame and the emissions probe in the experiments, estimated using volumetric flowrate and combustor volume (~0.02 s). To evaluate the impact of flame properties on combustion efficiency, the flow residence time in the PSR that matches experimentally determined combustion efficiency is calculated and indicated in the dashed line.

Since the experiment determines CO_2 mole fraction after subtracting the H_2O vapor in exhaust gas, the species mole fractions in the modeling results are reevaluated after removing H_2O mole fraction. Figure 11 compares experimental results with modeling results for CO_2 (top), O_2 (middle), and combustion efficiency (bottom). Here, the combustion efficiency is based on the measured/predicted CO_2 mole fraction normalized by CO_2 mole fraction in equilibrium. Except for the C15 cases, the modeling results from a single PSR capture the overall trends in experiments reasonably well, with the decrease of CO_2 mole fraction and the increase of O_2 mole fraction with increasing H_2 content. In the C15 cases (0, 20, 40% H_2), experimental results show considerably lower CO_2 mole fractions and higher O_2 mole fraction than the predicted values in the simulation, indicating incomplete combustion. Consequently, the combustion efficiencies based on CO_2 mole fraction in experiments exhibit significantly low values (~0.88) for C15 cases (0, 20, 40% H_2), compared to the model predictions, while exhibiting reasonable

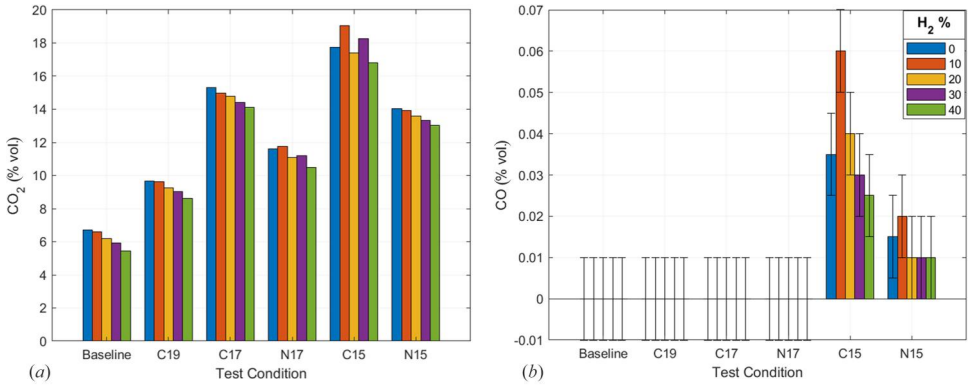

Fig. 7 Measurements of (*a*) CO_2 and (*b*) CO for all conditions, with results grouped by test case

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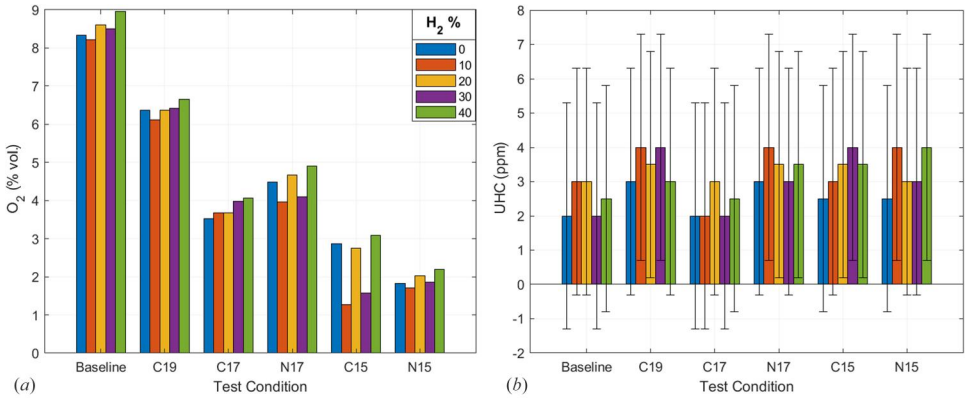

Fig. 8 Measurements of (*a*) O_2 and (*b*) UHC emissions for all conditions, with results grouped by test case

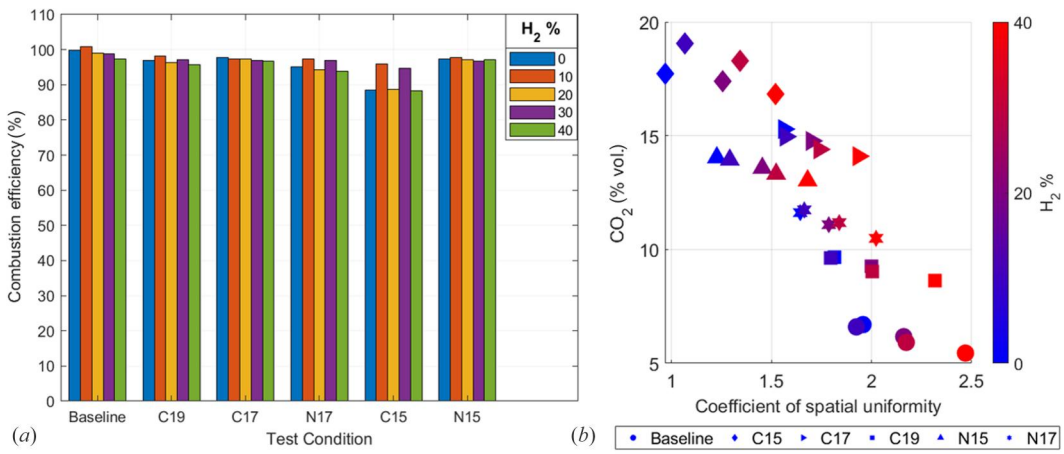

Fig. 9 Measurements of (*a*) combustion efficiency with results grouped by test case and (*b*) measured CO_2 as a function of the flame distribution coefficient for all conditions, with H_2 content indicated by the colorbar

agreements for 10 and 30% H_2. As discussed above, this somewhat inconsistency can be attributed to the flame stability specific to C15 cases.

To further analyze the results of C15 cases, the flow residence time in PSR calculation is reevaluated to the condition where the predicted combustion efficiency matches experimental results.

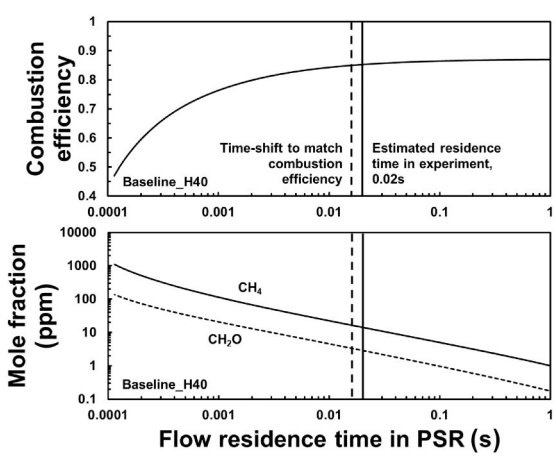

Fig. 10 Example PSR simulation for the baseline condition (Baseline_H40), exhibiting combustion efficiency, CH_4 and CH_2O mole fractions as a function of flow residence time

Figure 12 shows the reevaluated flow residence time normalized with the residence time at extinction limit, thus the Damköhler number (Da). While other cases have significantly large Da values for matching combustion efficiencies, regardless of the variation of H_2 content, the C15 cases (0, 20, 40% H_2) shows Da values close to unity, suggesting that the combustion processes occur near extinction limit. Interestingly, the N15 cases exhibit noticeably high *Da* values in general despite substantially reduced O_2 content, suggesting a unique contribution of CO_2 in the diluents to combustion efficiency. Both N15 and C15 cases exhibit relatively slower laminar flame speeds compared to other O_2 content cases (Table 1), thus slower reactivity and longer flow residence time at extinction limit in the PSR calculation. C15 has slightly slower reactivities (laminar flame speeds) than the N15 cases. This trend matches with the flame images shown in Fig. 6, where the flame/reaction zones are noticeably broader than in the other cases. While adding more H_2 content results in compact flame shapes as shown in Fig. 6, the low combustion efficiencies for C15 can be attributed to a noticeably slower rate constant of the major CO oxidation reaction, $CO + OH = CO_2 + H$. The modeling results suggest that the heat release is mainly produced by $H + O_2\ (+M) = HO_2\ (+M)$, $HO_2 + OH = H_2O + O_2$, and $CO + OH = CO_2 + H$. The rate constant of $CO + OH = CO_2 + H$ reaction is roughly two-order of magnitudes slower than those of the other two reactions at temperatures above 1000 K. Considering the rapid decrease in combustion efficiency near the extinction limit as shown in Fig. 11, the results shown in Fig. 12 indicate the need for employing a chemical reactor network approach, which would require careful

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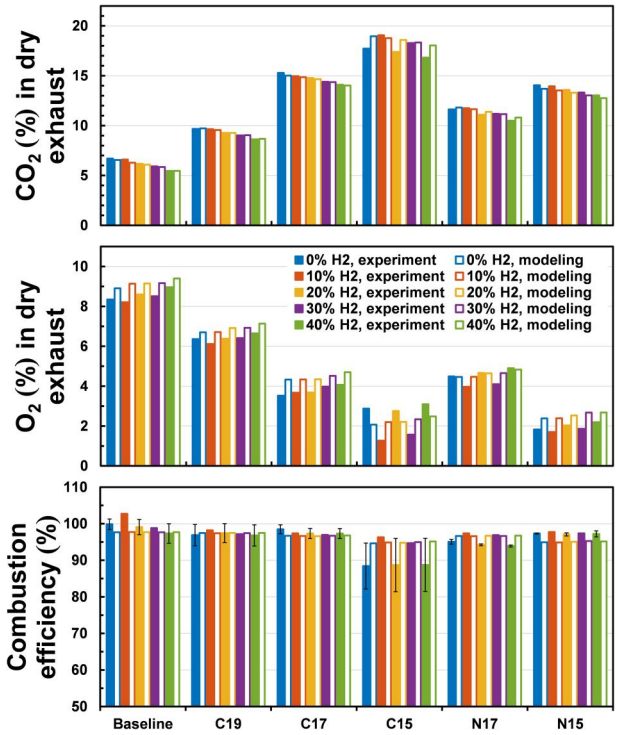

Fig. 11 Comparison between experimental and modeling results for CO_2 (top), O_2 (middle), and combustion efficiency (bottom). Flow residence time of 0.02 s is used for modeling results.

consideration for the impact of flame properties on the overall flow/flame structures and thus the overall time scales specific to each initial mixture composition.

To identify the potential unique contribution of CO_2 in the diluents over N_2 (C15 versus N15 cases), a series of flux analyses were performed, primarily focusing on key radicals (e.g., OH, H) and CO oxidation. The CO consumption pathway is dominated by a single reaction, $CO + OH = CO_2 + H$ (>99% in total consumption flux), regardless of flow residence time, as expected, suggesting the importance of OH for CO oxidation and thus combustion efficiency. At the flow residence time of 0.02 s, two major reactions play dominant roles in OH production, $H + O_2 = O + OH$ and $H_2O + O = OH + OH$, both chain branching reactions responsible for >80% of total OH production, regardless of the H_2, CO_2, and N_2 content in the initial mixtures. Figure 13 also shows that the overall OH consumption pathways are not much affected by the initial diluent compositions, except suggesting a strong impact by increasing H_2 content. While the variations in CO_2 and N_2 contents at low H_2 conditions do not affect overall OH consumption pathways, the increase in H_2 content augments the OH consumption channel through $H_2 + OH = H + H_2O$, producing the H atom, which is then recycled to produce OH by the reaction of $H + O_2 = O + OH$ (a major OH production pathway), which has a very strong temperature dependence.

On the other hand, CH_4 and CH_2O oxidation contribute considerable OH consumption flux. The increase of CO_2 content results in slightly lower reactor temperature compared to N_2 content, thus resulting in the increase of PSR flow residence time at the extinction limit, which can be attributed to higher specific heat value for CO_2 over N_2. The slightly reduced temperature in the C15 case results in slightly slower production of OH by temperature sensitive reaction over the N15 case, thus relatively lower combustion efficiency.

Although not shown in the figure, the OH mole fraction rapidly decreases near the extinction limit in the PSR. Major consumption channels for CH_4 and CH_2O oxidation are by the reaction with OH, thus the rapid decrease of OH mole fraction near extinction limit results in the rapid increases of CH_4 and CH_2O in exhaust gas, as shown in Fig. 10. Regardless of the initial mixture composition, the modeling results suggest that CH_4 and CH_2O are the major species in UHC (>80% of total UHC mole fraction). At 0.02 s flow residence time, the average mole fraction of CH_4 in dry exhaust gas is found to be 21 ppm with 2.7 ppm standard deviation, and that of CH_2O to be 4 ppm with 0.46 ppm standard deviation for all tested conditions. When the flow residence time is matched with the experimentally determined combustion efficiency, the average mole fractions of CH_4 and CH_2O in dry exhaust gas, except for the C15 cases (0, 20, 40% H_2) are found to be 17 ppm and 3.4 ppm, respectively. In the C15 cases (0, 20, 40% H_2), where combustion efficiencies are significantly lower at ~0.88, the modeling results suggest average values of 143 ppm CH_4 and 21 ppm CH_2O as UHC in dry exhaust gas. While CH_4 in UHC is well known as methane slip in natural gas combustion, the CH_2O in UHC needs further attention, particularly considering its notorious health impact.

Encouraged by the fact that the overall trend of the measured combustion efficiency and exhaust gas composition in an atmospheric condition is reasonably well captured by single PSR modeling, further numerical analysis at elevated pressures (up to 20 atm) was performed. Although not shown in the figure, the numerical result suggests an increase in combustion efficiency with pressure, which can be attributed to augmenting overall reaction rates due to the increase of the number density of molecules at high pressure, thus increasing temperature. On the other hand, the increase of pressure results in the substantial reduction of radical pool concentrations (OH, O and H), which leads to the reduction of NO_x emissions. In Zel'dovich thermal NO_x formation mechanism,

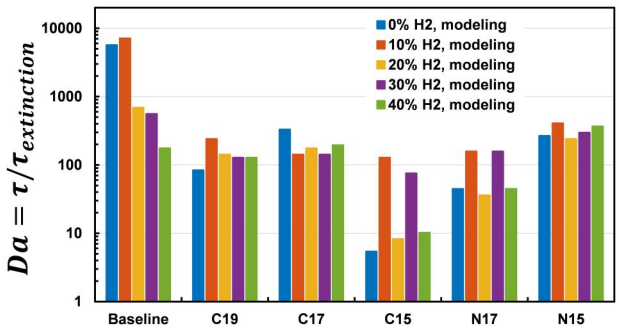

Fig. 12 Comparison of Damköhler number (Da) defined by the flow residence time matching experimentally determined combustion efficiency normalized by the flow residence time at extinction limit in PSR calculation

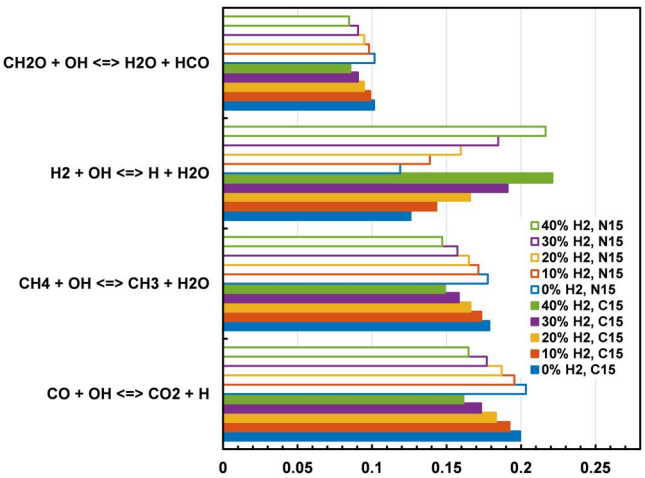

Fig. 13 Consumption pathways of OH radical at C15 and N15 cases at the flow residence time of 0.02 s in PSR calculations

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NO is initially produced from $N_2 + O = NO + N$ and $N + O_2 = NO + O$. In PSR calculations, NO is majorly consumed through $NO + O + M = NO_2 + M$ (~50%) and produced by $H + NO_2 = NO + OH$ (~50%) at all pressure conditions, known as NO–NO_2 conversion. Therefore, O and H play an important role in NO and NO_2 formation through thermal NO_x formation reactions. Consequently, the reduction of radical pool concentrations suppresses the overall formation of NO_x emissions. For example, Baseline 0H at 1 atm suggests 21 ppm NO_x emissions in dry exhaust gas, while 14 ppm at 20 atm, thus ~33% of reduction. Nevertheless, the current modeling approach based on single PSR might not reflect flame chemistry behaviors adequately at various pressure conditions, where the flame chemistry will be strongly coupled with molecular and turbulence transport characteristics

4 Conclusion

This study considers the effect of simulated EGR and hydrogen enrichment on the emissions performance of a natural gas swirl-stabilized flame. The test matrix includes various levels and compositions of EGR, using CO_2 and N_2 mixed with air, and hydrogen blends up to 40% by volume in natural gas. The combustion efficiency is noticeably affected by the highly diluted conditions, with CH* chemiluminescence imaging showing the change in flame shape as the reactant stream is modified. The addition of high levels of diluents results in an elongated, diffuse, and evenly distributed flame that exhibits variability in its combustion efficiency with different diluent compositions. Additionally, the emission measurements show an increase in CO for the highly diluted conditions. Companion chemical kinetic analysis identified critical reaction pathways that drive differences in the combustion efficiency, particularly the formation and consumption of the OH radical, which is highly dependent on the level of H_2 in the fuel. The use of EGR with fuel-flexible combustors can pose challenges in terms of both emissions and flame behavior that must be further addressed.

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Nomenclature

EGR = exhaust gas recirculation
L_{comb} = combustor length (cm)
P'_{RMS} = pressure fluctuation amplitude (Pa)
P_{atm} = atmospheric pressure (Pa)
$S_{L,max}$ = maximum stretched flame speed (cm/s)
S_L^o = unstretched laminar flame speed (cm/s)
T_{ad} = adiabatic flame temperature (° C)
x_a = volume fraction of a
κ_{ext} = extinction strain rate (1/s)
ϕ = equivalence ratio

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